DDN and POMDP Synthesis

Block 2 Day 5

Semester 1, 2026–2027 — Teaching Week 4

Session Overview

Session objective: Synthesize temporal belief, decision structure, and partially observed planning across healthcare and other applied domains.
Timing target (min): 72
Topic anchors: dynamic bayesian network; dynamic decision network; partially observable markov decision process; healthcare; belief update; human-in-the-loop
Padlet board: https://padlet.com/qmul/b2-d5-2xwxr9vzgkizrx5z

Exercises

Q1 Core Concept Check

Question type	Working mode	Expected time
Concept	Pair	10 min

Prompt

In a healthcare decision-support pipeline, what distinct job would you give to a dynamic Bayesian network (DBN), a dynamic decision network (DDN), and a partially observable Markov decision process (POMDP)? Add one sentence explaining how the three pieces connect.

Task details

- **Type:** concept
- **Mode:** pair
- **Time:** 10 min
- **Deliverable:** model-role mapping
- **Assessment focus:** map the roles of dynamic Bayesian networks, dynamic decision networks, and partially observable Markov decision processes
- **Expected length:** 3 rows + 1 link sentence

How to submit

Post one pair Padlet comment as a three-row table plus one linking sentence.

Discussion in class

Which model is hardest to replace with a simpler one?

Why this helps

Useful for synthesis questions that separate temporal state estimation, decision structure, and partial observability.

References

- Russell & Norvig (2021) AIMA 4e, Ch.17-18
- Poole & Mackworth (2023) 3e, Ch.12-13, pp.517-608

Q2 Structured Problem

Question type Problem → **Working mode** Individual → **Expected time** 13 min

Prompt

A patient has prior high-risk belief 0.25 at time t. The transition model is P(High_t+1|High_t)=0.70 and P(High_t+1|Low_t)=0.15. The observation model is P(Alert+|High)=0.80 and P(Alert+|Low)=0.30. After observing Alert+, compute the posterior belief that the patient is high risk at time t+1.

Formula reminder

$$b'(s') = \alpha\, P(o \mid s') \sum_{s} P(s' \mid s, a)\, b(s)$$

Use this when the task asks for a filtered or belief-state update.

Task details

- **Type:** problem
- **Mode:** individual
- **Time:** 13 min
- **Deliverable:** belief update computation
- **Assessment focus:** compute a prediction step and posterior belief after a noisy observation
- **Expected length:** 5 calculation lines

How to submit

Post your own Padlet comment showing the prediction step, the evidence update, the normalization step, and the final posterior.

Discussion in class

Why does the positive alert increase belief without making the patient certainly high risk?

Why this helps

Direct practice for belief-state updates while using different numbers from the exam-style examples.

References

- Russell & Norvig (2021) AIMA 4e, Ch.17-18
- Poole & Mackworth (2023) 3e, Ch.12-13, pp.517-608

Q3 Structured Problem

Question type	Working mode	Expected time
Problem	Individual	11 min

Prompt

Give two similarities and two differences between a dynamic decision network and a POMDP. Keep the answer in simple language and focus on what each model makes easier to express.

Task details

- **Type:** problem
- **Mode:** individual
- **Time:** 11 min
- **Deliverable:** model comparison note
- **Assessment focus:** compare dynamic decision networks and POMDPs without treating them as synonyms
- **Expected length:** 4 bullets

How to submit

Post your own Padlet comment as four bullets.

Discussion in class

Which model makes the role of belief state more explicit?

Why this helps

Useful for comparison questions that test conceptual understanding rather than only notation.

References

- Russell & Norvig (2021) AIMA 4e, Ch.17-18
- Poole & Mackworth (2023) 3e, Ch.12-13, pp.517-608

Q4 Collaborative Mini-Case

Question type	Working mode	Expected time
Case	Group	14 min

Prompt

Design an end-to-end healthcare decision-support workflow using a data agent, a state-estimation agent, a policy agent, and a safety or human-oversight gate. Include the points where evidence, belief, recommendation, and approval appear.

Task details

- **Type:** case
- **Mode:** group
- **Time:** 14 min
- **Deliverable:** workflow design

- **Assessment focus:** design an end-to-end healthcare workflow with human and safety gates
- **Expected length:** 6 steps

How to submit

Post one group Padlet comment with the role order, the handoff points, and one place where a human must approve or override the recommendation.

Discussion in class

Which stage should be allowed to recommend but never act automatically?

Why this helps

Supports capstone case questions on integrated uncertainty reasoning and human-in-the-loop design.

References

- Russell & Norvig (2021) AIMA 4e, Ch.17-18
- Poole & Mackworth (2023) 3e, Ch.12-13, pp.517-608

Q5 Exam-Style Constructed Response

Question type **Exam** → **Working mode** **Pair** → **Expected time** **14 min**

Prompt

Using healthcare as your reference point, explain how the same hidden-state, noisy-evidence, belief-update, and action-choice pattern could also appear in autonomous driving, financial risk, or cybersecurity. Your answer should make clear what transfers and what stays domain-specific.

Task details

- **Type:** exam
- **Mode:** pair
- **Time:** 14 min
- **Deliverable:** structured synthesis response
- **Assessment focus:** explain how the same uncertain-decision pattern transfers across domains
- **Expected length:** 190-250 words

How to submit

Post one pair Padlet comment comparing healthcare with one other domain and identifying what stays the same and what must change.

Discussion in class

Which part transfers well across domains, and which part stays domain-specific?

Why this helps

Supports capstone-style preparation through a fresh applied scenario.

References

- Russell & Norvig (2021) AIMA 4e, Ch.17-18
- Poole & Mackworth (2023) 3e, Ch.12-13, pp.517-608

Q6 Stretch Extension

Question type Stretch → **Working mode** Group → **Expected time** 10 min

Prompt

Define a compact monitoring dashboard for the healthcare capstone deployment. Cover four headings: clinical quality, operational load, safety, and trust. Give at least one metric for each heading.

Task details

- **Type:** stretch
- **Mode:** group
- **Time:** 10 min
- **Deliverable:** monitoring dashboard
- **Assessment focus:** define a compact deployment dashboard for the capstone setting
- **Expected length:** 4 categories + 1 metric each

How to submit

Post one group Padlet comment with the four headings and one metric under each.

Discussion in class

Which single metric would you watch first during the first live week?

Why this helps

Good extension for evaluation, monitoring, and trustworthy deployment questions.

References

- Russell & Norvig (2021) AIMA 4e, Ch.17-18
- Poole & Mackworth (2023) 3e, Ch.12-13, pp.517-608

Submission Instructions

Use the seeded Padlet posts for Q1–Q6. Q2 and Q3 are individual. Q1 and Q5 are pair tasks. Q4 and Q6 should be one joint group comment. Start each comment with your names or group label.

Whole-Class Debrief Prompts

- Which of DBN, DDN, and POMDP plays the clearest role in the pipeline?
- Where should the human gate sit in a high-risk workflow?
- What changes most when you move the same pattern into another domain?

Exam Transfer Note (Day-Level)

Maps to capstone exam preparation on DBN, DDN, POMDP, belief update, workflow design, and cross-domain transfer under uncertainty.